

Beyond Mimicry: Toward Lifelong Adaptability in Imitation Learning

Blue Sky Ideas Track — AAMAS 2026

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Motivation

A child stacking blocks

A child watches an adult stack blocks.

They do **not** memorise every hand motion.

Instead, they extract **reusable primitives**:

- *Grasp, lift, place*

Later, with blocks of different shapes and colours, the child builds structures **never seen before**.



Key question

Why can't our imitation learning agents do the same?

Imitation learning today

Modern IL methods achieve impressive **trajectory fidelity**:

- Behavioural Cloning (BC), high accuracy on training distributions
- Inverse Reinforcement Learning, infers underlying objectives
- Adversarial IL, handles imperfect demonstrations

But...

When the context shifts beyond the training distribution, **performance collapses**.
These failures are not edge cases, they are the norm.

Root cause: IL has been optimised for *sample efficiency*, not *adaptability*.
No amount of memorisation enables behaviour in genuinely novel contexts.

The Problem with Mimicry

Two forms of imitation

Mimicry

- Retrieve the most similar training state
- Replay the associated action
- Leads to *behaviour-seeking mode*
- Failure modes:
 - Compounding errors
 - Memorisation

Both forms matter

Mimicry works in controlled, static environments.

Adaptive behaviour is required when **variation is the norm**.

Adaptive Behaviour

- Understand *why*, not just *what*
- Decompose demonstrations into reusable primitives
- Recombine for novel contexts
- Near-optimal on unseen states

Why current evaluation hides the problem

Standard metric: **Average Episodic Reward (AER)**

Problems with AER:

- Averaging hides catastrophic failures on novel states
- An agent can score 90% by memorising familiar trajectories while failing everywhere else
- Cannot distinguish *trajectory memorisation* from *compositional understanding*
- States not seen during training create **hidden failure points**: invisible at evaluation, catastrophic in deployment

We need metrics that directly measure generalisation.

Compositional Repertoire Learning

Compositional Repertoire Learning (CRL)

Thesis: Redefine success in IL from *trajectory reproduction* to *compositional generalisation*.

Demonstrations contain two layers of information:

1. **Surface trajectories:** what current methods capture
2. **Deep compositional structure:** what CRL targets

CRL requires agents to:

- Extract **behavioural primitives** (sub-goal achieving steps)
- Learn **compositional rules** governing their combination
- Recombine them for **novel contexts without retraining**

This shift is not merely technical — it is philosophical.

It changes what it *means* for an agent to imitate successfully.

Goal-conditioned Contextual MDPs (GCMDPs)

Standard MDPs conflate task definition with dynamics — impossible to isolate what varies compositionally.

GCMDPs extend contextual MDPs with three key additions:

1. **Explicit goals:** trajectory-level success criteria (terminal, landmark, ordered landmark)
When an agent fails, we know which primitive or rule broke down.
2. **Controllable contexts:** expose generative parameters (e.g., block count, map layout)
Systematic variation: change one factor at a time.
3. **Deterministic transitions:** eliminate stochasticity as a confound
Failures reflect agent limitations, not bad luck.

Block-stacking example

Contexts control: number of blocks, block colours, stacking order.

Goals specify: which configuration counts as success.

Measuring compositional generalisation

Three dimensions (from linguistics, applied to behaviour):

- **Systematicity**: recombine known primitives in new structural configurations
- **Productivity**: extend to longer sequences / greater complexity than seen in training
- **Substitutivity**: replace components with functional equivalents (e.g., new colour block)

The Generalisation Boundary $\mathcal{B}_\pi(p)$:

- Measures the maximum contextual distance at which agent maintains performance $\geq p$
- Uses **Levenshtein distance** over symbolic context vectors
- Two agents with identical accuracy can have **vastly different boundaries**

Example

Agent A handles substitutivity up to distance 3 but fails systematicity at distance 1.
Agent B shows the opposite. Same AER. Completely different capabilities.

Research Agenda

CRL is a **research agenda**, not a single algorithm. Key open directions:

1. **Hybrid architectures**

Foundation models provide semantic priors; planners handle long-horizon reasoning.
Learn primitives via imitation, compose them via principled algorithms.

2. **Multi-agent compositional IL**

Teams share and recombine primitives: collective repertoire expansion via social learning.
Mirrors cultural transmission in human societies.

3. **Cognitive science synergies**

Developmental psychology: structured progression from mimicry to composition.
Kolb's experiential learning: reflect, extract structure, test combinations.

4. **Compositional safety**

Safety constraints must cover entire *combination spaces*, not individual trajectories.

Conclusion

1. **The problem:** IL is optimised for mimicry; evaluation hides generalisation failures.
2. **The shift:** From trajectory reproduction → compositional adaptability.
3. **The framework:** GCMDPs + compositional dimensions + generalisation boundary metric.
4. **The agenda:** Hybrid architectures, multi-agent IL, cognitive science, safety.

The vision

Agents that learn behavioural primitives **once** and recombine them **everywhere** without retraining, without memorisation, without brittleness.

*Imitation learning can be the cornerstone of next-generation AI,
but only if we stop measuring the wrong thing.*

Thank you!

Questions?

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Imitation Learning • Goal Recognition
Lifelong Learning • Multi-Agent Systems

Contact



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A *Goal-conditioned Contextual MDP* is a tuple $\mathcal{M}_{gc} = \langle \mathcal{S}, \mathcal{A}, \mathcal{C}, \mathcal{G}, \mathcal{O}, \mathcal{T}, \phi, d \rangle$, where:

- $\mathcal{S}$ — state space; $\mathcal{A}$ — action space
- $\mathcal{C}$ — set of controllable contexts
- $\mathcal{G}$ — set of goals (terminal / landmark / ordered landmark)
- $\mathcal{O}$ — observation space
- $\mathcal{T} : \mathcal{O} \times \mathcal{A} \mapsto \mathcal{O}$ — deterministic transition
- $\phi : \mathcal{S} \times \mathcal{C} \mapsto \mathcal{O}$ — projection function
- $d : \mathcal{C} \times \mathcal{C} \rightarrow \mathbb{R}_{\geq 0}$ — contextual distance metric

Backup: How is CRL different from hierarchical / skill-based IL?

Hierarchical and skill-based IL (e.g., options framework, GAIL variants):

- Learn reusable behavioural components — yes
- Typically assume **fixed composition rules**
- Do not measure whether recombination actually generalises

CRL goes further:

- Explicitly measures compositional recombination *across contexts*
- Provides a metric (generalisation boundary) to compare agents on *how well* they compose
- Requires benchmarks with controllable contexts and isolated compositional dimensions